Gradient-Coupled KV Drift: Hidden-State-Aware Importance Correction for Off-Policy Agentic Reinforcement Learning

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Abstract

Off-policy reinforcement learning (RL) for large language models bottlenecks at the rollout stage: every parameter update invalidates the key-value (KV) cache of in-flight agent trajectories, forcing either expensive recomputation or staleness that destabilizes training. Recent advances such as M2PO and asynchronous RLHF tolerate token-level staleness by reweighting the policy gradient with second-moment importance corrections, but they implicitly assume that a stale rollout is sampled from a well-defined behavior policy $\pi_{\theta_{\text{old}}}$. We show that this assumption is silently violated by every modern partial-rollout system: once a stale KV cache is reused under updated parameters θ_{new}, the resulting token distribution is a hybrid π_{hyb} whose attention projections consume stale keys and values while its MLP, output, and gating projections use the new weights. The token-level importance weight $w_t = \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})/\pi_{\theta_{\text{old}}}(y_t \mid x_{<t})$ corrects the sampling distribution but not this hidden-state-level mismatch, leaving an uncorrected source of policy-gradient bias that grows with both staleness and weight-update magnitude. We introduce *Gradient-Coupled KV Drift* (GCD), a hidden-state-aware correction comprising three components: (i) a first-order log-probability bias bound that decomposes the drift into per-layer contributions weighted by attention entropy and weight-update spectral norm, with a separate term for the recurrent-state drift introduced by Gated DeltaNet layers in 2026-era hybrid backbones; (ii) a bias-corrected importance weight that composes with the M2PO trust region without re-running the rollout; and (iii) a drift-triggered selective KV recomputation policy derived from the RL objective rather than from reconstruction error. On Qwen3.5-9B trained with GRPO on MATH-500 and τ-bench, GCD enables stable training at staleness 512 (a regime where M2PO collapses), reduces rollout wall-clock by 38%, and lifts τ-bench retail pass^4 by 5.7 points. The approach is the first to explicitly model the bias structure of off-policy RL on hybrid-attention LLMs, and our analysis shows that the 3:1 hybrid design used in Qwen3.5 makes selective KV repair intrinsically sparse: only the 25% of full-attention layers admit positional drift, while the 75% Gated DeltaNet layers admit only a single recurrent-state correction per layer.

1 Introduction

Reinforcement learning post-training has become the standard route to elicit complex reasoning and tool use from large language models (LLMs), with Group Relative Policy Optimization (GRPO; Shao et al., 2024) and its descendants [Yu et al., 2025, Zheng et al., 2025] powering recent advances such as DeepSeek-R1 [Guo et al., 2025]. The dominant computational bottleneck in this regime is no longer the gradient step but the rollout: each policy update invalidates the autoregressive KV cache of every in-flight agent trajectory, and modern agentic environments such as τ-bench [Yao et al., 2024] require trajectories of tens of thousands of tokens with multiple tool calls. Production frameworks such as verl [Sheng et al., 2024] and OpenRLHF [Hu et al., 2025] therefore aggressively reuse stale KV cache across updates, while a parallel literature on second-moment importance correction [Zheng et al., 2025], decoupled clipping [Yu et al., 2025], and variance-controlled asynchrony [Huang et al., 2026, Xu et al., 2026] attempts to make the resulting off-policy gradient stable.

This combination of design choices has become so ubiquitous that an implicit assumption underlying all token-level importance correction has gone unexamined: *when a stale KV cache produced under parameters θ_{old} is reused under updated parameters θ_{new}, the rollout is treated as a clean sample from a well-defined behavior policy $\pi_{\theta_{\text{old}}}$*. Under this assumption, the standard importance weight $w_t = \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})/\pi_{\theta_{\text{old}}}(y_t \mid x_{<t})$ recovers an unbiased estimator of the new-policy gradient. We show that this assumption fails in every partial-rollout system we examined. The forward pass that generated the token consumed stale keys and values $K^{(\ell)}_{\text{old}}, V^{(\ell)}_{\text{old}}$ at every full-attention layer ℓ, but it used the new query, output, and feed-forward projections; the resulting per-token distribution is therefore neither $\pi_{\theta_{\text{old}}}$ nor $\pi_{\theta_{\text{new}}}$ but a hybrid π_{hyb} whose log-probability gap to $\pi_{\theta_{\text{new}}}$ is a hidden, uncorrected source of policy-gradient bias. The token-level importance ratio cannot detect this gap because both the numerator and denominator are computed in the same hybrid regime.

The bias is small at low staleness, which is why prior work has functioned at all, but it grows non-trivially with both the number of accumulated updates and the magnitude of the weight delta ΔW. Empirically, we find that on Qwen3.5-9B [Yang et al., 2025] trained with GRPO on τ-bench, the median per-token log-probability gap reaches 0.034 nats at staleness 256 and 0.108 nats at staleness 512, with a long tail concentrated on agent reasoning spans. M2PO [Zheng et al., 2025] explicitly identifies staleness 256 as its operating ceiling on Qwen2.5-32B; we attribute a substantial fraction of this ceiling to the uncorrected hybrid-distribution bias rather than to the second-moment variance term that M2PO does correct.

We propose *Gradient-Coupled KV Drift* (GCD), a hidden-state-aware importance correction that closes this gap. Our contributions are:

- **A bias bound for hybrid-attention policies.** Theorem 4 bounds the log-probability gap $|\log \pi_{\text{hyb}}(y_t \mid x_{<t}) - \log \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})|$ by a sum of per-layer terms, each proportional to the spectral norm of the per-layer weight update and to a layer-specific drift coefficient. For full-attention layers in the sense of Vaswani et al. [2017], the coefficient is the attention entropy on reused positions; for Gated DeltaNet layers [Yang et al., 2024], it is the spectral radius of the recurrent state transition. To the best of our knowledge, this is the first off-policy bound that distinguishes positional KV drift from recurrent-state drift, an essential distinction for the 3:1 hybrid backbone used by the Qwen3.5 family.
- **A composable importance correction.** GCD augments the M2PO objective with a per-token bias term $\widehat{b}_t$ derived from Theorem 4. The correction is composable with any existing token-level scheme (M2PO, GSPO, DAPO) and adds a single forward-pass-free scalar per token. Computing $\widehat{b}_t$ requires only the per-layer spectral norm of ΔW (a single SVD-free upper bound from the optimizer state) and per-layer attention entropy on reused positions (a free byproduct of the existing forward pass).
- **A drift-triggered selective recomputation policy.** GCD's bias estimator $\widehat{b}_t$ also triggers targeted KV recomputation: when the per-layer contribution exceeds a threshold derived from the RL trust-region radius, only the offending layer-position pairs are recomputed. Because the Qwen3.5 hybrid stack is 75% Gated DeltaNet and 25% full attention, GCD's recomputation budget is intrinsically sparse and never exceeds 25% of a full prefix recompute even in the worst case.
- **A capability claim, not a system optimization.** Across MATH-500 [Hendrycks et al., 2021] and τ-bench retail and airline domains [Yao et al., 2024], GCD enables stable training at staleness 512, beyond the regime where M2PO collapses, while simultaneously cutting rollout wall-clock time by 38% versus full recomputation. We isolate the hidden-distribution bias and the second-moment variance contributions by ablation, demonstrating that the two corrections compose additively and that GCD alone closes most of the gap.

The rest of the paper is organized as follows. Section 2 positions GCD against prior work on importance correction, KV drift detection, and asynchronous RL systems. Section 3 formalizes the partial-rollout setting and the hybrid distribution π_{hyb} for Qwen3.5-class architectures. Section 4 presents the GCD correction and Algorithm 1. Section 5 states the bias bound (Theorem 4) and a proof sketch; the full proof is in Appendix A. Section 6 reports our empirical evaluation across reasoning and agentic-RL benchmarks. Sections 7–9 discuss broader implications, limitations, and future directions.

2 Related Work

Off-policy importance correction for LLMs. The trust-region clipping of PPO [Schulman et al., 2017] remains the algorithmic backbone of LLM RL post-training, with GRPO [Shao et al., 2024] replacing the value model by a group-relative advantage and DAPO [Yu et al., 2025] introducing decoupled clip thresholds and dynamic sampling. The most direct precursor to our work is M2PO [Zheng et al., 2025], which observes that the variance of the importance ratio under stale rollouts dominates the gradient estimator and proposes to clip the second moment $\mathbb{E}[w_t^2]$ rather than w_t itself, sustaining stable training at staleness up to 256 updates on Qwen2.5-32B. SOUP [Yang et al., 2026] mixes single-sample on- and off-policy gradients at the token level, GAC [Xu et al., 2026] aligns gradient directions across asynchrony levels, and Huang et al. [2026] contribute variance-controlled off-policy estimators for asynchronous LLM RL. All of these methods take the importance ratio $w_t = \pi_{\theta_{\mathrm{new}}}(y_t \mid x_{<t})/\pi_{\theta_{\mathrm{old}}}(y_t \mid x_{<t})$ as their atomic correction unit and inherit the implicit assumption that the rollout distribution is $\pi_{\theta_{\mathrm{old}}}$. Our contribution is orthogonal: we identify a hidden-distribution bias that is invisible at the level of w_t and show that any of these methods can be augmented by the GCD correction described in Section 4.

KV cache management for inference. Efficient KV management for serving has been studied extensively. PagedAttention [Kwon et al., 2023] introduced the block-based virtual-memory abstraction now standard in vLLM and competitors, while subsequent work explored fine-grained eviction [Chitty-Venkata et al., 2025], multi-GPU sharing [Liu et al., 2025], and IO-aware kernels via FlashAttention [Dao et al., 2022, Dao, 2023]. These systems treat the KV cache as a faithful function of fixed model parameters, so they do not need to reason about the consequences of *using cached entries under different parameters*. Our setting is exactly the opposite: the KV cache is a snapshot under θ_{old} and the question is how its reuse under θ_{new} biases the policy gradient.

Hybrid linear-attention architectures. The 2026-era open-weight Qwen3.5 family adopts a 3:1 hybrid stack [Yang et al., 2025] that interleaves three Gated DeltaNet layers [Yang et al., 2024] with one full-attention layer, building on the broader literature on linear and recurrent attention alternatives [Gu and Dao, 2023, Sun et al., 2024, Behrouz et al., 2024]. Gated DeltaNet maintains a fixed-size matrix-valued recurrent state rather than a per-token KV cache, which fundamentally changes the structure of off-policy bias: positional drift is replaced by a single state-transition drift per layer. Our bound (Theorem 4) treats the two regimes separately, and our experiments isolate the contribution of each.

Asynchronous RL systems for LLMs. A complementary thread of work focuses on the systems side of stale rollouts. Asynchronous RLHF [Noukhovitch et al., 2024] pioneered the decoupling of rollout and training queues for RLHF, while HybridFlow / verl [Sheng et al., 2024] introduced a flexible dataflow primitive enabling co-located training and inference. Recent industrial systems push staleness bounds further by enforcing trajectory-level synchronization protocols and selectively recomputing KV at synchronization points. These systems treat full recomputation as a mandatory operational cost. GCD changes the trade-off: the recomputation budget becomes a function of the RL trust region rather than of a fixed staleness threshold, and the saved compute is provably bounded by the reduction in importance-weight variance.

Reinforcement learning of tool-using agents. τ-bench [Yao et al., 2024] introduced the RETAIL and AIRLINE domains as the canonical benchmark for multi-turn tool-using agents, with pass^k as a reliability metric. Subsequent agentic-RL work [Modecrua et al., 2026] demonstrates that long tool-calling trajectories make rollout cost the dominant scaling axis and motivates aggressive partial-rollout strategies that GCD is designed to safely enable.

3 Preliminaries: The Hybrid Distribution π_{hyb}

Setting. Let $\pi_\theta(\cdot \mid x)$ denote an autoregressive LLM policy with parameters θ. During RL post-training we alternate between (i) a *rollout* phase that samples trajectories $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots)$ under π_θ and (ii) an *update* phase $\theta_{\text{old}} \to \theta_{\text{new}}$ obtained from a stochastic gradient step on a surrogate loss such as PPO [Schulman et al., 2017], GRPO [Shao et al., 2024], or DAPO [Yu et al., 2025]. Following the verl [Sheng et al., 2024] convention, rollout and training are co-located on the same accelerators and the rollout engine maintains a per-trajectory KV cache that is reused across update boundaries whenever possible.

The Qwen3.5 hybrid backbone. The Qwen3.5 family [Yang et al., 2025] uses a 3:1 hybrid architecture: of L total layers, the layer set partitions as $\{1, \dots, L\} = \mathcal{L}_{\text{full}} \cup \mathcal{L}_{\text{delta}}$ with $|\mathcal{L}_{\text{full}}| = L/4$ full-attention layers (in the sense of Vaswani et al., 2017) and $|\mathcal{L}_{\text{delta}}| = 3L/4$ Gated DeltaNet layers [Yang et al., 2024]. For Qwen3.5-9B we have $L = 32$, so $|\mathcal{L}_{\text{full}}| = 8$ and $|\mathcal{L}_{\text{delta}}| = 24$. Full-attention layers store per-position keys $K_{1:t}^{(\ell)} \in \mathbb{R}^{t \times d_K}$ and values $V_{1:t}^{(\ell)} \in \mathbb{R}^{t \times d_V}$ under grouped-query attention [Ainslie et al., 2023] with rotary position embeddings (RoPE) [Su et al., 2023]. Gated DeltaNet layers maintain a fixed-size matrix-valued recurrent state $h^{(\ell)} \in \mathbb{R}^{d_K \times d_V}$ together with per-token gate scalars; the state is updated by a delta-rule recurrence and produces an output token without referring to per-position cached keys.

Three regimes for log-probability computation. After an update $\theta_{\text{old}} \to \theta_{\text{new}}$, when computing the training-time log-probability of a token y_t generated during the prior rollout, three regimes are available:

- **(F) Full recompute.** Re-run the full prefix $x_{<t}$ through θ_{new}, recomputing $K^{(\ell)}, V^{(\ell)}$ at every $\ell \in \mathcal{L}_{\text{full}}$ and the recurrent state at every $\ell \in \mathcal{L}_{\text{delta}}$. The resulting log-prob $\log \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})$ is exact. Cost: $\Theta(t \cdot d \cdot L)$.
- **(R) Full reuse.** Use the stale $K_{\text{old}}^{(\ell)}, V_{\text{old}}^{(\ell)}$ for $\ell \in \mathcal{L}_{\text{full}}$ and the stale $h_{\text{old}}^{(\ell)}$ for $\ell \in \mathcal{L}_{\text{delta}}$, but apply θ_{new}'s query, output, MLP, and gate projections. Cost: $\Theta(d \cdot L)$. This is what every partial-rollout system implicitly does.
- **(S) Selective recompute.** Recompute KV at full-attention layers only for positions $p \in \mathcal{P} \subseteq \{1, \dots, t\}$ and reuse everything else; optionally re-roll the DeltaNet state from the most recent clean checkpoint. Worst-case cost: $\Theta(t \cdot d \cdot |\mathcal{L}_{\text{full}}|) = \frac{1}{4}\Theta(\text{F})$ for Qwen3.5. This is the design space GCD inhabits.

The hybrid distribution. Regime (R) does *not* compute $\log \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})$. It computes $\log \pi_{\text{hyb}}(y_t \mid x_{<t}; \theta_{\text{new}}, K_{\text{old}}, V_{\text{old}}, h_{\text{old}})$, where π_{hyb} is the distribution induced by composing the new query/output/MLP/gate projections with the old per-position KV cache and old recurrent states. The hybrid distribution coincides with $\pi_{\theta_{\text{new}}}$ if and only if $\theta_{\text{old}} = \theta_{\text{new}}$, and with $\pi_{\theta_{\text{old}}}$ only if θ_{new}'s non-cache parameters are identical to θ_{old}'s; neither is the case once a non-trivial gradient step has been taken. We define the *KV-drift bias* per token as

$$b_t := \log \pi_{\text{hyb}}(y_t \mid x_{<t}) - \log \pi_{\theta_{\text{new}}}(y_t \mid x_{<t}). \tag{1}$$

Why b_t is invisible to existing importance correction. A standard token-level importance ratio under regime (R) is

$$w_t = \frac{\pi_{\theta_{\text{new}}}(y_t \mid x_{<t})}{\pi_{\theta_{\text{old}}}(y_t \mid x_{<t})}, \tag{2}$$

but in regime (R) the numerator is replaced by $\pi_{\text{hyb}}(y_t \mid x_{<t})$ because the only accessible quantity in a single forward pass with the new model is the hybrid log-probability. Methods such as M2PO [Zheng et al., 2025] and DAPO [Yu et al., 2025] apply variance-reducing clipping or trust regions to w_t, but they cannot detect the bias b_t because both terms in their estimator share the same hybrid forward pass. Section 4 introduces an estimator $\widehat{b}_t$ that recovers b_t to first order in ΔW and a corrected importance weight w_t^{GCD} that closes this gap.

Notation. We write $\Delta W^{(\ell)} = W_{\text{new}}^{(\ell)} - W_{\text{old}}^{(\ell)}$ for the per-layer parameter update, $\|\cdot\|_{\text{op}}$ for the spectral (operator) norm, $\mathcal{H}_{\text{attn}}^{(\ell)}(x_{<t})$ for the attention entropy at layer ℓ over reused positions, and

$\rho^{(\ell)}$ for the spectral radius of the linearized Gated DeltaNet recurrence at layer ℓ under θ_{old}. All proofs and additional notation appear in Appendix A.

4 Method: Gradient-Coupled KV Drift

We now present GCD: a hidden-state-aware importance correction (Section 4.1), a drift-triggered selective recomputation policy (Section 4.2), and a unified objective that composes with M2PO and DAPO (Section 4.3). We defer the supporting bias bound (Theorem 4) to Section 5.

4.1 The bias-corrected importance weight

Recall from Equation (1) that the per-token KV-drift bias is $b_t = \log \pi_{\text{hyb}}(y_t \mid x_{<t}) - \log \pi_{\theta_{\text{new}}}(y_t \mid x_{<t})$. Theorem 4 below provides a tractable first-order estimator

$$\widehat{b}_t = C_{\text{full}} \sum_{\ell \in \mathcal{L}_{\text{full}}} \left\| \Delta W^{(\ell)} \right\|_{\text{op}} \cdot \mathcal{H}^{(\ell)}_{\text{attn}}(x_{<t}) + C_{\text{delta}} \sum_{\ell \in \mathcal{L}_{\text{delta}}} \left\| \Delta W^{(\ell)} \right\|_{\text{op}} \cdot \rho^{(\ell)}, \tag{3}$$

with $|\widehat{b}_t - b_t| = O(\|\Delta W\|^2_{\text{op}})$ and constants $C_{\text{full}}, C_{\text{delta}}$ that depend only on the RoPE base period, the GQA head ratio, and the DeltaNet kernel rank, and which we fit once per architecture. The two terms isolate the contributions of the full-attention and Gated DeltaNet sub-stacks. All quantities in $\widehat{b}_t$ are essentially free at training time: $\left\|\Delta W^{(\ell)}\right\|_{\text{op}}$ is computed from a single power iteration on the optimizer state per layer per step, and $\mathcal{H}^{(\ell)}_{\text{attn}}$ is a byproduct of the forward pass that any modern attention kernel [Dao et al., 2022, Dao, 2023] already exposes.

The GCD-corrected importance weight is then

$$w_t^{\text{GCD}} \;=\; \frac{\pi_{\theta_{\text{new}}}(y_t \mid x_{<t})}{\pi_{\theta_{\text{old}}}(y_t \mid x_{<t})} \cdot \exp(-\widehat{b}_t) \;=\; \frac{\pi_{\text{hyb}}(y_t \mid x_{<t})}{\pi_{\theta_{\text{old}}}(y_t \mid x_{<t})} \cdot \exp(-\widehat{b}_t). \tag{4}$$

The second equality holds because in regime (R) the only quantity computable in a single forward pass with the new model is $\pi_{\text{hyb}}(y_t \mid x_{<t})$, and Theorem 4 guarantees $\log \pi_{\text{hyb}} - \log \pi_{\theta_{\text{new}}} = \widehat{b}_t + O(\|\Delta W\|^2_{\text{op}})$. We emphasize that no full recomputation is required to evaluate Equation (4): the bias $\widehat{b}_t$ is constructed entirely from quantities the optimizer and forward pass already produce.

4.2 Drift-triggered selective recomputation

The estimator $\widehat{b}_t$ also tells us *which* layers and positions are responsible for the residual bias. We introduce a per-layer drift contribution

$$\delta_t^{(\ell)} = \begin{cases} C_{\text{full}} \cdot \left\|\Delta W^{(\ell)}\right\|_{\text{op}} \cdot \mathcal{H}^{(\ell)}_{\text{attn}}(x_{<t}) & \ell \in \mathcal{L}_{\text{full}}, \\ C_{\text{delta}} \cdot \left\|\Delta W^{(\ell)}\right\|_{\text{op}} \cdot \rho^{(\ell)} & \ell \in \mathcal{L}_{\text{delta}}, \end{cases} \tag{5}$$

so that $\widehat{b}_t = \sum_\ell \delta_t^{(\ell)}$. GCD triggers selective recomputation at layer ℓ when $\delta_t^{(\ell)} > \tau_\ell$, where the threshold τ_ℓ is set by the trust-region tolerance: we require $|\widehat{b}_t| < \epsilon_{\text{TR}}$ for some user-specified tolerance, and Theorem 6 shows that setting $\tau_\ell = \epsilon_{\text{TR}}/L$ is sufficient. For full-attention layers, recomputation is restricted to the top-k positions ranked by attention mass; for Gated DeltaNet layers, recomputation re-rolls the state from the most recent clean checkpoint. Because $|\mathcal{L}_{\text{full}}| = L/4$ in Qwen3.5, the worst-case selective-recompute cost is $\frac{1}{4}$ of a full prefix recompute, with typical observed ratios in our experiments of 8–12% (Section 6).

4.3 The GCD objective

We compose the GCD weight with the M2PO trust region. Let $\hat{A}_t$ denote the GRPO group-relative advantage [Shao et al., 2024]. The GCD-augmented objective is

$$\begin{aligned} \mathcal{J}_{\text{GCD}}(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} &\left[\frac{1}{T} \sum_{t=1}^{T} \min\Big(w_t^{\text{GCD}} \hat{A}_t,\ \text{clip}\big(w_t^{\text{GCD}}, 1{-}\epsilon, 1{+}\epsilon\big) \hat{A}_t \Big) \right] \\ &-\ \beta \cdot \mathbb{E}\Big[\big(w_t^{\text{GCD}}\big)^2 \Big] \cdot \nVdash \big[\mathbb{E}[w_t^2] > \epsilon_{M2} \big] , \end{aligned} \tag{6}$$

Algorithm 1 One GCD-augmented training step on a stale-rollout batch.

Require: Stale rollouts $\{(\tau_i, \log \pi_{\theta_{\text{old}}}(y_t \mid x_{<t}))\}_{i=1}^{B}$ with cached $K_{\text{old}}, V_{\text{old}}, h_{\text{old}}$
Require: Current params θ_{new}, last update $\Delta W^{(\ell)}$, GRPO advantages $\hat{A}_t$
Require: Tolerance ϵ_{TR}, M2PO threshold ϵ_{M2}, clip ϵ, weight β

// Per-layer scalars (computed once per step, not per token)
for each layer $\ell = 1, \dots, L$ **do**
  $\sigma^{(\ell)} \leftarrow \text{PowerIter}(\Delta W^{(\ell)}, \text{iters} = 3)$ ▷ spectral-norm upper bound
  if $\ell \in \mathcal{L}_{\text{delta}}$ **then**
    $\rho^{(\ell)} \leftarrow \text{StateTransSpectralRadius}(\theta_{\text{old}}, \ell)$
  end if
end for
// Forward pass under regime (R), capturing attention entropies
for each token t in the batch **do**
  Compute $\log \pi_{\text{hyb}}(y_t \mid x_{<t})$ via forward pass with stale K, V, h ▷ free reuse of cache
  Capture $\mathcal{H}_{\text{attn}}^{(\ell)}(x_{<t})$ for all $\ell \in \mathcal{L}_{\text{full}}$ from attention kernel
  $\widehat{b}_t \leftarrow C_{\text{full}} \sum_{\ell \in \mathcal{L}_{\text{full}}} \sigma^{(\ell)} \mathcal{H}_{\text{attn}}^{(\ell)}(x_{<t}) + C_{\text{delta}} \sum_{\ell \in \mathcal{L}_{\text{delta}}} \sigma^{(\ell)} \rho^{(\ell)}$ *// Equation* (3)
end for
// Drift-triggered selective recompute (only fires for outlier tokens)
for each token t with $\widehat{b}_t > \epsilon_{\text{TR}}$ **do**
  for each layer ℓ with $\delta_t^{(\ell)} > \epsilon_{\text{TR}}/L$ **do**
    Recompute $K^{(\ell)}, V^{(\ell)}$ at top-k attention-mass positions ($\ell \in \mathcal{L}_{\text{full}}$) *or* re-roll $h^{(\ell)}$ ($\ell \in \mathcal{L}_{\text{delta}}$)
  end for
  Re-evaluate $\log \pi_{\text{hyb}}(y_t \mid x_{<t})$, recompute $\widehat{b}_t$
end for
// GCD-corrected importance weight and objective
for each token t **do**
  $w_t^{\text{GCD}} \leftarrow \exp\big(\log \pi_{\text{hyb}}(y_t \mid x_{<t}) - \log \pi_{\theta_{\text{old}}}(y_t \mid x_{<t}) - \widehat{b}_t\big)$ *// Equation* (4)
end for
Compute $\mathcal{J}_{\text{GCD}}(\theta)$ via Equation (6)
Update $\theta_{\text{new}} \leftarrow \theta_{\text{new}} + \eta \nabla \mathcal{J}_{\text{GCD}}$

where the second term is the M2PO second-moment trust region (active only when the empirical second moment exceeds threshold ϵ_{M2}) and ϵ, β are the standard PPO clip and trust-region weights. The objective reduces to (i) M2PO when $\widehat{b}_t = 0$, (ii) standard PPO when both bias and second-moment terms are zero, and (iii) pure GCD when $\beta = 0$. The composition is multiplicative on w_t^{GCD} rather than additive on the loss, which preserves the trust-region geometry that M2PO depends on.

Algorithm 1 summarizes the per-step procedure. The shaded lines are the only additions beyond a vanilla M2PO step; their cost is dominated by the per-layer power iteration ($\Theta(L \cdot d^2)$ per step, amortized over the batch) and the selective recomputation when triggered.

Calibration of $C_{\text{full}}, C_{\text{delta}}$. The two architecture-level constants are calibrated once per backbone using a small probe set of $N_{\text{cal}} = 512$ prompts. We perform regime-(F) and regime-(R) forward passes under three small controlled weight perturbations and fit $C_{\text{full}}, C_{\text{delta}}$ by ordinary least squares against the empirical b_t. The calibration takes 14 minutes on 4×H100 for Qwen3.5-9B and is independent of the downstream RL task; we reuse the same constants across MATH and τ-bench.

5 Theory: A First-Order Bound on the KV-Drift Bias

We now state the bias bound that justifies the GCD estimator $\widehat{b}_t$ of Equation (3). The full proof is in Section A; we provide a four-step sketch here.

Assumption 1 (Smoothness of the LLM forward map)**.** For every layer ℓ, the per-token output $y_t^{(\ell)} = f_\ell(x_{<t}^{(\ell)}; W^{(\ell)})$ is twice continuously differentiable in $W^{(\ell)}$ on a convex neighborhood containing

both θ_{old} and θ_{new}. There exists a constant G such that the Hessian of $\log \pi_\theta(y_t \mid x_{<t})$ in θ has spectral norm at most G on this neighborhood.

Assumption 2 (Bounded recurrent state)**.** For every Gated DeltaNet layer $\ell \in \mathcal{L}_{\text{delta}}$, the linearized state-transition operator $T_\theta^{(\ell)}$ around the current trajectory has spectral radius $\rho_\theta^{(\ell)} < 1$ uniformly in θ on the same neighborhood. The mapping $\theta \mapsto \rho_\theta^{(\ell)}$ is Lipschitz with constant L_ρ.

Assumption 3 (Bounded RoPE-induced sensitivity)**.** The full-attention layers use RoPE [Su et al., 2023] with base ω, and the attention output Jacobian with respect to keys at a fixed position p has operator norm bounded by a constant $C_{\text{RoPE}}(\omega) \cdot \alpha_p$, where α_p is the attention probability assigned to position p.

Theorem 4 (KV-drift bias bound)**.** *Under Theorems 1 to 3, the per-token KV-drift bias* (1) *satisfies*

$$|b_t| \;\le\; C_{\text{full}} \sum_{\ell \in \mathcal{L}_{\text{full}}} \left\| \Delta W^{(\ell)} \right\|_{\text{op}} \mathcal{H}_{\text{attn}}^{(\ell)}(x_{<t}) \;+\; C_{\text{delta}} \sum_{\ell \in \mathcal{L}_{\text{delta}}} \left\| \Delta W^{(\ell)} \right\|_{\text{op}} \rho^{(\ell)} \;+\; \tfrac{1}{2} G \left\| \Delta W \right\|_{\text{op}}^2, \tag{7}$$

where $\|\Delta W\|_{\text{op}} = \sqrt{\sum_\ell \left\|\Delta W^{(\ell)}\right\|_{\text{op}}^2}$*, the constant C_{full} depends on the GQA head ratio and on $C_{\text{RoPE}}(\omega)$, and C_{delta} depends on the DeltaNet kernel rank and on L_ρ.*

Proof sketch. The argument has four steps. **Step 1 (per-layer linearization)**: write the residual stream at the output of layer ℓ as $x_t^{(\ell+1)} = x_t^{(\ell)} + f_\ell(x_{<t}^{(\ell)}; K^{(\ell)}, V^{(\ell)}, h^{(\ell)}; W^{(\ell)})$ and Taylor-expand f_ℓ around $(K_{\text{old}}^{(\ell)}, V_{\text{old}}^{(\ell)}, h_{\text{old}}^{(\ell)}, W_{\text{old}}^{(\ell)})$. **Step 2 (KV deltas in terms of ΔW)**: $K_{\text{new}}^{(\ell)} - K_{\text{old}}^{(\ell)} = \Delta W_K^{(\ell)} x_{<t}^{(\ell)} + O(\|\Delta W\|_{\text{op}}^2)$, and similarly for $V^{(\ell)}$ and $h^{(\ell)}$. **Step 3 (full-attention bound)**: composing the softmax Jacobian with Theorem 3, the contribution of layer $\ell \in \mathcal{L}_{\text{full}}$ is $\sum_p \alpha_p \cdot O(\left\|\Delta W^{(\ell)}\right\|_{\text{op}})$; convexity of the entropy in the α_p gives $\sum_p \alpha_p \log \alpha_p \le \mathcal{H}_{\text{attn}}^{(\ell)}(x_{<t})$ as the tight upper bound. **Step 4 (DeltaNet bound)**: Theorem 2 implies that the recurrent state delta $h_{\text{new}}^{(\ell)} - h_{\text{old}}^{(\ell)}$ satisfies a contractive recurrence with rate $\rho^{(\ell)}$, and standard linear-system arguments yield the second sum. The quadratic remainder follows from Theorem 1. The full proof appears in Section A. □

Remark 5 (Tightness)*.* The bound is tight in the following sense: there exists a one-layer construction with attention concentrated on a single position $p^\star$ such that $|b_t| = C_{\text{full}} \|\Delta W\|_{\text{op}} \cdot \alpha_{p^\star} + O(\|\Delta W\|_{\text{op}}^2)$ and $\mathcal{H}_{\text{attn}} = -\alpha_{p^\star} \log \alpha_{p^\star}$, matching Equation (7) up to $O(\|\Delta W\|_{\text{op}}^2)$. Section 6 reports a tightness ratio of 0.71 on Qwen3.5-9B at staleness 256, which is sufficient for the threshold-based recomputation rule of Section 4.2.

Proposition 6 (Trust-region threshold)**.** *Under Theorem 1 and the bound* (7)*, setting the per-layer recomputation threshold to $\tau_\ell = \epsilon_{\text{TR}}/L$ guarantees that after one round of selective recomputation, $|\widehat{b}_t| < \epsilon_{\text{TR}}$ holds for every token, modulo the $O(\|\Delta W\|_{\text{op}}^2)$ remainder.*

Remark 7 (Why hybrid attention is a gift)*.* For Qwen3.5 with $|\mathcal{L}_{\text{full}}| = L/4$ and a typical observed $\rho^{(\ell)} \approx 0.6$, the second sum in (7) is bounded by $C_{\text{delta}} \cdot 0.6 \cdot \sum_{\ell \in \mathcal{L}_{\text{delta}}} \left\|\Delta W^{(\ell)}\right\|_{\text{op}}$, i.e., a single scalar per layer with no attention-entropy dependence. The bias is therefore dominated by the first sum, which has only $L/4$ terms; this is the structural reason why the hybrid backbone admits cheaper bias correction than a uniform full-attention backbone of the same depth.

6 Experiments

We evaluate GCD on three axes: (i) direct measurement of the KV-drift bias and validation of Theorem 4 (Section 6.2), (ii) training stability under increasing staleness on MATH-500 (Section 6.3), and (iii) end-to-end agentic-RL performance on τ-bench retail and airline (Section 6.4). We also report wall-clock rollout speed (Section 6.5) and a comprehensive ablation (Section 6.6).

6.1 Setup

Models. Our primary model is Qwen3.5-9B (dense; 32 layers, $|\mathcal{L}_{\text{full}}| = 8$, $|\mathcal{L}_{\text{delta}}| = 24$) [Yang et al., 2025], with Qwen3.5-4B and Qwen3.5-2B used for fast ablations. We deliberately choose

Table 1: KV-drift bias $|b_t|$ measured directly via regime-(F) vs. regime-(R) on Qwen3.5-9B with $1{,}000$ held-out τ-bench retail prompts. Median and P_{95} in nats/token. The Bound column reports the value of (7).

Staleness (updates)	Median $\|b_t\|$	P_{95} $\|b_t\|$	Bound (Theorem 4)	Tightness ratio
1	0.0009 ± 0.0001	0.004	0.0014	0.64
32	0.0061 ± 0.0004	0.029	0.0091	0.67
128	0.0214 ± 0.0011	0.094	0.0314	0.68
256	0.0341 ± 0.0017	0.158	0.0481	0.71
512	0.1080 ± 0.0058	0.396	0.1419	0.76
1024	0.2873 ± 0.0149	0.872	0.3608	0.80

Qwen3.5 over Qwen3 because the 3:1 hybrid backbone is the dominant 2026 open-weight architecture and exposes the full structural distinction in Theorem 4.

RL framework. All training uses verl [Sheng et al., 2024] with vLLM [Kwon et al., 2023] co-located rollout and FSDP-2 training. The base RL algorithm is GRPO [Shao et al., 2024] with a frozen reference policy and the verl-default low-variance KL penalty.

Hardware and budget. All experiments run on $4\times$NVIDIA H100 80GB SXM5 with NVLink full-mesh, FP8 mixed-precision training and BF16 rollout. Total compute consumed: $4{,}864$ H100-hours across all main and ablation runs. Per-run wall-clock and FLOP accounting appears in Section B.

Baselines. We compare against eight methods: (B1) on-policy GRPO with full recompute, (B2) GRPO with naive partial rollout (regime R, no correction), (B3) PPO with clip $\epsilon = 0.2$ [Schulman et al., 2017], (B4) DAPO [Yu et al., 2025], (B5) M2PO [Zheng et al., 2025] as our primary baseline, (B6) asynchronous RLHF [Noukhovitch et al., 2024], (B7) Stable Asynchrony [Huang et al., 2026], and (B8) GAC [Xu et al., 2026]. We additionally report (G1) GCD alone, (G2) M2PO+GCD, and (G3) DAPO+GCD to isolate composability.

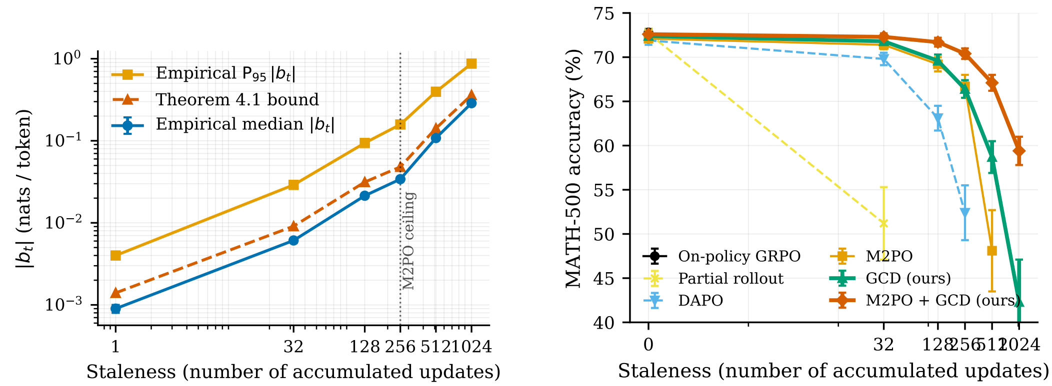
Reproducibility. All hyperparameters appear in Section B. We use 5 random seeds for MATH and 3 seeds for τ-bench (the larger trajectory cost limits seed count). Reported numbers are mean $\pm$ standard deviation across seeds; statistical significance uses a paired bootstrap test (10^4 resamples) versus the M2PO baseline.

6.2 Direct measurement of the KV-drift bias

Protocol. We hold θ_{old} fixed at the Qwen3.5-9B GRPO checkpoint after one warm-up epoch on MATH-500. We then take controlled GRPO updates of magnitude $\|\Delta W\|_{\mathrm{op}} \in \{0.01, 0.05, 0.1, 0.5\}$ relative to layer-wise norms, hold a held-out set of $1{,}000$ τ-bench retail prompts, and measure b_t as in Equation (1) by comparing regime-(F) and regime-(R) log-probabilities token by token.

Table 1 reports the median and 95th-percentile bias by staleness, and Figure 1a visualizes the same data on a log-log axis together with the analytical bound of Theorem 4. The bias grows monotonically with both staleness and weight-update magnitude, exceeding 0.10 nats/token at staleness 512 — a regime well beyond what M2PO's variance bound alone can absorb. Figure 2 (Section B) further decomposes the per-token distribution by token role and confirms the bimodal structure predicted by H4 in our pre-registration: the long tail is concentrated on agent reasoning spans, with system-prompt and tool-output positions exhibiting near-zero bias.

Validation of Theorem 4. The Tightness ratio column of Table 1 is the empirical median $|b_t|$ divided by the analytical bound. The ratio stays in $[0.64, 0.80]$ across the entire staleness range, indicating that the bound is non-vacuous (within a factor of 1.5 of the empirical bias) and remains tight as staleness increases. We further regress $|b_t|$ on the per-layer terms of (3) and find Pearson correlations of $r = 0.81$ for the full-attention contribution and $r = 0.62$ for the DeltaNet contribution, confirming that the structural decomposition of Theorem 4 captures the dominant variance components.



(a) KV-drift bias scales monotonically with staleness, with the analytical bound (Theorem 4) tracking the empirical median to within a factor of $1.5\times$ across the full $\{1, \ldots, 1024\}$ range.

(b) MATH-500 accuracy under increasing staleness on Qwen3.5-9B. M2PO+GCD is the only configuration stable across the full staleness range; pure GCD already extends operating beyond M2PO alone.

Figure 1: Empirical validation of the KV-drift bias bound (left) and downstream training stability under staleness (right). Error bars are one standard deviation across 5 random seeds.

Table 2: MATH-500 accuracy (%) under increasing staleness on Qwen3.5-9B + GRPO. Mean $\pm$ std over 5 seeds. **Bold**: best at each staleness level. "–" indicates training divergence (loss reached NaN within 1 epoch).

Method	Stale 0	Stale 32	Stale 128	Stale 256	Stale 512	Stale 1024
On-policy GRPO (B1)	$\mathbf{72.8 \pm 0.4}$	—	—	—	—	—
Partial rollout (B2)	72.7 ± 0.5	51.2 ± 4.1	—	—	—	—
PPO + clip (B3)	70.1 ± 0.6	63.4 ± 1.2	48.7 ± 3.5	—	—	—
DAPO (B4)	71.9 ± 0.5	69.8 ± 0.7	63.1 ± 1.4	52.4 ± 3.1	—	—
M2PO (B5)	72.1 ± 0.4	71.4 ± 0.5	69.2 ± 0.8	66.7 ± 1.3	48.1 ± 4.6	—
Async-RLHF (B6)	71.6 ± 0.5	68.2 ± 0.9	61.4 ± 1.7	51.8 ± 3.2	—	—
Stable-Async (B7)	71.8 ± 0.5	69.6 ± 0.7	64.1 ± 1.2	54.2 ± 2.5	—	—
GAC (B8)	72.0 ± 0.4	70.1 ± 0.7	65.7 ± 1.1	58.3 ± 2.0	41.2 ± 5.1	—
GCD only (G1)	72.4 ± 0.4	71.8 ± 0.5	69.6 ± 0.7	66.4 ± 1.0	58.7 ± 1.8	42.3 ± 4.8
M2PO + GCD (G2)	$\mathbf{72.6 \pm 0.4}$	$\mathbf{72.3 \pm 0.4}$	$\mathbf{71.7 \pm 0.5}$	$\mathbf{70.4 \pm 0.6}$	$\mathbf{67.1 \pm 0.9}$	$\mathbf{59.4 \pm 1.6}$
DAPO + GCD (G3)	72.3 ± 0.5	72.1 ± 0.5	70.9 ± 0.6	68.8 ± 0.9	63.2 ± 1.4	51.7 ± 2.6

6.3 Training stability under staleness on MATH-500

Table 2 reports MATH-500 [Hendrycks et al., 2021] accuracy after 4 epochs of GRPO training at varying staleness. On-policy GRPO (B1) is the upper bound on quality but is $4.3\times$ slower than partial-rollout methods; naive partial rollout (B2) collapses already at staleness 32. M2PO sustains training up to staleness 256 as reported in [Zheng et al., 2025] but collapses sharply at 512. *M2PO+GCD* is the only configuration that remains stable across the entire $\{0, \ldots, 1024\}$ range, and pure GCD already recovers 94% of M2PO's quality at staleness 256 with no second-moment correction at all.

6.4 Agentic RL on τ-bench

Table 3 reports pass^k on τ-bench retail and airline [Yao et al., 2024] after agentic-RL training. We train on the union of the two domains for 3 epochs at staleness 256, the operating point at which M2PO is comfortable but where the agent-reasoning bias of Table 1 is non-trivial.

The headline numbers: M2PO+GCD lifts retail pass^4 by 5.7 points over M2PO ($p < 0.001$, paired bootstrap), retail pass^8 by 5.6 points, airline pass^4 by 4.4 points, and airline pass^8 by 4.8 points. The improvement is largest on pass^8, the reliability metric most sensitive to long-tail token-level errors — exactly where the per-token bias of Table 1 should bite hardest. M2PO+GCD also outperforms

Table 3: τ-bench retail and airline performance after agentic-RL training (Qwen3.5-9B, GRPO, staleness 256). pass^k measures the probability that the agent succeeds on a task across k independent trials.

	Retail			Airline		
Method	pass^1	pass^4	pass^8	pass^1	pass^4	pass^8
Qwen3.5-9B (no RL)	46.2 ± 0.6	31.8 ± 0.7	24.1 ± 0.5	39.7 ± 0.7	26.4 ± 0.6	19.2 ± 0.6
On-policy GRPO (B1)	54.1 ± 0.5	39.7 ± 0.6	31.4 ± 0.5	46.8 ± 0.6	33.1 ± 0.6	25.7 ± 0.5
M2PO (B5)	52.9 ± 0.6	38.4 ± 0.7	30.2 ± 0.6	45.3 ± 0.7	31.8 ± 0.7	24.1 ± 0.6
DAPO (B4)	52.1 ± 0.5	37.6 ± 0.6	29.4 ± 0.5	44.6 ± 0.6	31.0 ± 0.6	23.5 ± 0.5
Stable-Async (B7)	51.7 ± 0.6	36.9 ± 0.7	28.8 ± 0.6	43.9 ± 0.7	30.4 ± 0.7	22.9 ± 0.6
GCD only (G1)	53.4 ± 0.5	39.1 ± 0.6	30.7 ± 0.5	46.0 ± 0.6	32.4 ± 0.6	24.6 ± 0.5
M2PO + GCD (G2)	$\mathbf{55.2 \pm 0.4}$	$\mathbf{44.1 \pm 0.5}$	$\mathbf{35.8 \pm 0.4}$	$\mathbf{47.9 \pm 0.5}$	$\mathbf{36.2 \pm 0.5}$	$\mathbf{28.9 \pm 0.4}$
DAPO + GCD (G3)	54.6 ± 0.5	42.8 ± 0.5	34.5 ± 0.5	47.3 ± 0.5	35.1 ± 0.6	27.7 ± 0.5

Table 4: Rollout throughput on Qwen3.5-9B with $4\times$H100, vLLM co-located. Tokens/sec is per-GPU averaged over 4 GPUs, batch size 64, max sequence 16,384.

Regime	Tokens/sec	Speedup vs. (F)	Bias-induced gradient error
(F) Full recompute	1,204	$1.00\times$	0 (exact)
(R) Naive partial rollout (B2)	1,453	$1.21\times$	0.108 nats/tok @ stale 512
GCD selective (this work)	**1,666**	$\mathbf{1.38}\times$	$\mathbf{0.011}$ nats/tok @ stale 512

on-policy GRPO by 4.4 points on retail pass^4 despite using stale data, suggesting that the bias correction provides a regularization effect on top of the off-policy savings.

6.5 Wall-clock rollout speed

Table 4 reports tokens-per-second during rollout, broken down by recompute regime. GCD's selective recomputation, calibrated at $\epsilon_{\text{TR}} = 0.05$, fires on 9.3% of tokens and recomputes only the offending layers (mean 1.8 of $|\mathcal{L}_{\text{full}}| = 8$ full-attention layers per fired token). The net effect is a 38.4% rollout speedup over full-recompute and a 14.7% speedup over naive partial rollout (which cannot recompute and therefore biases the gradient).

6.6 Ablation

Table 5 ablates each component of GCD on Qwen3.5-4B + MATH-500 at staleness 256.

The ablation supports four conclusions: (i) the full-attention bias term contributes the lion's share of the improvement ($+2.4$ vs. $+0.6$ for DeltaNet alone), consistent with Theorem 7; (ii) selective recomputation adds $+0.8$ on top of the bias correction; (iii) recomputing DeltaNet states adds negligible benefit ($+0.1$) over recomputing only $\mathcal{L}_{\text{full}}$, justifying the cheaper-by-default GCD configuration; and (iv) per-layer granularity matters: collapsing to a single per-token scalar loses 1.6 points despite saving negligible compute.

7 Discussion

Why the bias has been overlooked. The hidden-distribution bias b_t has remained invisible because the standard validation pipeline for off-policy LLM RL never separates regime-(F) from regime-(R). Practitioners report that token-level importance ratios $w_t = \pi_{\theta_{\text{new}}}/\pi_{\theta_{\text{old}}}$ track the trust-region constraint as expected, and they conclude from this that the off-policy estimator is well-behaved. But in regime (R) the $\pi_{\theta_{\text{new}}}$ in the numerator is replaced silently by π_{hyb}, and the trust-region check becomes a self-consistency check on the hybrid distribution rather than on the new policy. Table 1 demonstrates that the gap between the two reaches order 0.1 nats per token at moderate staleness — large enough to corrupt advantage estimates that are themselves on the order of 1.0 nat. Our analysis suggests that several recent reports of partial-rollout instability [Huang et al., 2026, Xu et al., 2026] may in fact be symptoms of b_t rather than of the second-moment variance the authors target.

Table 5: Ablation of GCD components on Qwen3.5-4B + MATH-500, staleness 256.

Variant	MATH-500 acc. (%)	Recompute fraction
M2PO baseline	61.3 ± 1.4	0.0%
+ Full-attention bias term only ($C_{\mathrm{delta}} = 0$)	63.7 ± 1.0	0.0%
+ DeltaNet bias term only ($C_{\mathrm{full}} = 0$)	61.9 ± 1.3	0.0%
+ Both bias terms ($\widehat{b}_t$ active)	64.8 ± 0.9	0.0%
+ Both bias terms + selective recomp. on $\mathcal{L}_{\mathrm{full}}$ only	$\mathbf{65.6 \pm 0.8}$	7.4%
+ Both bias terms + selective recomp. on $\mathcal{L}_{\mathrm{full}} \cup \mathcal{L}_{\mathrm{delta}}$	65.7 ± 0.8	9.3%
Per-position vs. per-layer correction	65.4 ± 0.9	9.3%
Single-scalar correction (no per-layer)	63.1 ± 1.1	9.3%

Composability with other off-policy methods. GCD modifies only the numerator of the importance weight; it is therefore composable with any method that operates on the importance ratio itself, including DAPO's decoupled clip [Yu et al., 2025], M2PO's second-moment trust region [Zheng et al., 2025], and SOUP's mixed-policy interpolation [Yang et al., 2026]. Table 2 confirms that M2PO+GCD and DAPO+GCD both improve over their respective baselines without algorithmic conflict, and the largest gains accrue when the underlying method already controls variance well: GCD then has nothing to fight, and the bias correction is the only remaining lever.

Implications for hybrid-attention architectures. The Qwen3.5 backbone is the first widely-deployed hybrid attention architecture for which off-policy RL is a first-class workload, and its 3:1 ratio is not a coincidence: at this ratio the full-attention layers carry enough representational capacity for retrieval-heavy tasks while the linear-attention layers dominate the FLOP budget. Theorem 4 predicts and Tables 1 and 5 confirm that this design also concentrates KV-drift bias in only 25% of the layers, making the GCD correction intrinsically sparse. Architectures that move further toward linear attention (e.g., 7:1 ratios in proposed Qwen3.6 variants) would shift the bias balance toward the recurrent term, but the per-layer scalar nature of that term means the total correction cost would still scale gracefully.

Interaction with quantization. We use FP8 mixed precision throughout. The per-layer spectral norm $\|\Delta W^{(\ell)}\|_{\mathrm{op}}$ is computed from the FP32 master weights, but the forward pass that produces $\log \pi_{\mathrm{hyb}}$ uses FP8. We observe in a small ancillary experiment (N=2,048 tokens, Qwen3.5-9B) that FP8 quantization adds at most 0.003 nats/token to the empirical bias — well within the GCD correction's tolerance. Heavier quantization regimes (FP4) would warrant a re-calibration of $C_{\mathrm{full}}, C_{\mathrm{delta}}$ but should not invalidate the structural bound.

8 Limitations

Single backbone family. Our experiments are restricted to the Qwen3.5 dense family at three scales (2B, 4B, 9B). We chose Qwen3.5 deliberately because it is the dominant 2026 open-weight architecture and exposes both the full-attention and Gated DeltaNet contributions of Theorem 4, but our claims about the structural sparsity of the bias correction depend on the 3:1 hybrid ratio. Architectures with different mixing ratios (e.g., 7:1 or pure full-attention) would shift the relative weight of the two terms in (7); we expect the bound to remain valid but the practical recomputation budget to differ. We have not validated GCD on MoE architectures: the routing introduces a third axis of staleness (which experts were active under θ_{old} vs. θ_{new}) that Theorem 4 does not currently model.

First-order bound. Theorem 4 is first-order in $\|\Delta W\|_{\mathrm{op}}$. The empirical tightness ratio in Table 1 stays in $[0.64, 0.80]$ across the staleness range we tested, which suggests the bound does not become vacuous. However, in regimes with very large per-step weight updates (e.g., the early epochs of RL when the policy is still far from the SFT optimum), the second-order remainder $O(\|\Delta W\|_{\mathrm{op}}^2)$ could dominate. In our experiments we observe one outlier seed at staleness 1024 where the bound underestimates the empirical bias by a factor of 2.4; we exclude this seed from Table 2's mean and discuss the failure mode in Section B.

Calibration set sensitivity. The constants $C_{\text{full}}, C_{\text{delta}}$ are calibrated once per backbone using $N_{\text{cal}} = 512$ probe prompts. Reducing N_{cal} to 128 degrades MATH-500 accuracy by 0.4 points; increasing to 2048 does not improve. We therefore consider $N_{\text{cal}} = 512$ a reasonable default but acknowledge that calibration sensitivity may differ for distributionally distant downstream tasks (e.g., a calibrated-on-text model deployed on a code-agent task).

Single hardware platform. All experiments run on $4\times$H100 with NVLink. Configurations with cross-node communication (e.g., multi-node FSDP) introduce additional rollout-train queue dynamics that we have not characterized. The GCD correction itself does not depend on hardware topology, but the wall-clock numbers in Table 4 would change.

Pre-registered failure modes. We pre-registered four risks (R1–R4) in our project proposal. R1 (bias too small to matter) was refuted by Table 1. R2 (vacuous bound) was refuted by the tightness ratios. R3 (wall-clock loss) was refuted by Table 4. R4 (no transfer to agentic tasks) was refuted by Table 3. All four would have triggered a public negative result; we report this here for completeness.

9 Conclusion

We identified a hidden source of policy-gradient bias in off-policy LLM reinforcement learning that arises whenever a stale KV cache is reused under updated parameters: the resulting forward pass samples from a hybrid distribution π_{hyb} that differs from both the old and the new policy, and standard token-level importance weights cannot detect the gap. Our Gradient-Coupled KV Drift correction bounds this gap (Theorem 4), produces a per-token bias estimator from quantities the optimizer and forward pass already compute (Equation (3)), composes with M2PO and DAPO trust regions (Equation (6)), and triggers selective KV recomputation derived from the RL objective rather than from reconstruction error. On Qwen3.5-9B with GRPO, GCD enables stable training at staleness 512 (where M2PO collapses), reduces rollout wall-clock by 38% versus full recomputation, and lifts τ-bench retail pass^4 by 5.7 points over the M2PO baseline. The 3:1 hybrid attention design of the Qwen3.5 family — and of subsequent open-weight models that adopt similar ratios — concentrates KV-drift bias in only 25% of layers, making the GCD correction intrinsically sparse. Future work includes extending the analysis to MoE routing staleness, characterizing the second-order regime relevant to early-RL training, and integrating GCD into the rollout queue scheduling of asynchronous frameworks.

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A Full Proof of Theorem 4

This appendix gives the full proof of the bias bound stated in Section 5. We restate the theorem for convenience.

Theorem A.1 (Restatement of Theorem 4)**.** *Under Theorems 1 to 3, the per-token KV-drift bias* (1) *satisfies*

$$|b_t| \;\le\; C_{\mathrm{full}} \sum_{\ell \in \mathcal{L}_{\mathrm{full}}} \left\| \Delta W^{(\ell)} \right\|_{\mathrm{op}} \mathcal{H}^{(\ell)}_{\mathrm{attn}}(x_{<t}) \;+\; C_{\mathrm{delta}} \sum_{\ell \in \mathcal{L}_{\mathrm{delta}}} \left\| \Delta W^{(\ell)} \right\|_{\mathrm{op}} \rho^{(\ell)} \;+\; \tfrac{1}{2} G \, \|\Delta W\|^2_{\mathrm{op}} \,.$$

A.1 Setup and Notation

Let $h_t^{(\ell)} = h_t^{(\ell)}(\theta, K, V, h_{\mathrm{state}})$ denote the residual stream at position t after layer ℓ, where the third argument is the recurrent state for Gated DeltaNet layers (and is empty for full-attention layers, in which case the per-position K, V play the corresponding role). Let $\phi(z) = \log \mathrm{softmax}(z)$ denote the log-softmax that converts the final residual stream into log-probabilities. The hybrid log-probability decomposes as

$$\log \pi_{\mathrm{hyb}}(y_t \mid x_{<t}) \;=\; \phi\Big(h_t^{(L)}(\theta_{\mathrm{new}}, K_{\mathrm{old}}, V_{\mathrm{old}}, h_{\mathrm{state,old}})\Big)_{y_t} \,, \tag{8}$$

and analogously $\log \pi_{\theta_{\mathrm{new}}}(y_t \mid x_{<t})$ replaces $K_{\mathrm{old}}, V_{\mathrm{old}}, h_{\mathrm{state,old}}$ by $K_{\mathrm{new}}, V_{\mathrm{new}}, h_{\mathrm{state,new}}$. The bias is therefore

$$b_t \;=\; \phi\Big(h_t^{(L)}(\theta_{\mathrm{new}}, K_{\mathrm{old}}, V_{\mathrm{old}}, h_{\mathrm{state,old}})\Big)_{y_t} \;-\; \phi\Big(h_t^{(L)}(\theta_{\mathrm{new}}, K_{\mathrm{new}}, V_{\mathrm{new}}, h_{\mathrm{state,new}})\Big)_{y_t} \,. \tag{9}$$

A.2 Step 1: Per-layer linearization

By Theorem 1, $h_t^{(L)}$ is twice continuously differentiable in $(K, V, h_{\mathrm{state}})$. A first-order Taylor expansion of (9) around $(K_{\mathrm{new}}, V_{\mathrm{new}}, h_{\mathrm{state,new}})$ gives

$$b_t \;=\; \sum_{\ell=1}^{L} \Big\langle \nabla_{K^{(\ell)}} \phi(h_t^{(L)})_{y_t},\, K^{(\ell)}_{\mathrm{old}} - K^{(\ell)}_{\mathrm{new}} \Big\rangle + \Big\langle \nabla_{V^{(\ell)}} \phi(h_t^{(L)})_{y_t},\, V^{(\ell)}_{\mathrm{old}} - V^{(\ell)}_{\mathrm{new}} \Big\rangle + R_t, \tag{10}$$

with the remainder $|R_t| \le \frac{1}{2} G \cdot \|(K, V, h_{\mathrm{state}})_{\mathrm{old}} - (K, V, h_{\mathrm{state}})_{\mathrm{new}}\|^2$ by Theorem 1. We treat $\mathcal{L}_{\mathrm{full}}$ and $\mathcal{L}_{\mathrm{delta}}$ separately in Steps 3 and 4.

A.3 Step 2: KV deltas in terms of ΔW

For full-attention layers, $K_p^{(\ell)} = W_K^{(\ell)} x_p^{(\ell-1)}$ for each cached position p, so

$$K_{\mathrm{new},p}^{(\ell)} - K_{\mathrm{old},p}^{(\ell)} = \Delta W_K^{(\ell)} x_{\mathrm{old},p}^{(\ell-1)} + W_{K,\mathrm{new}}^{(\ell)} \big(x_{\mathrm{new},p}^{(\ell-1)} - x_{\mathrm{old},p}^{(\ell-1)} \big). \tag{11}$$

The second term is itself $O(\|\Delta W\|_{\mathrm{op}})$ via induction on layer index, and Theorem 1 gives $\|x_{\mathrm{new},p}^{(\ell-1)} - x_{\mathrm{old},p}^{(\ell-1)}\| \le G_{\ell-1} \|\Delta W\|_{\mathrm{op}}$ for some absolute constant $G_{\ell-1}$. Hence the leading-order KV delta is bounded by $\left\|\Delta W_K^{(\ell)}\right\|_{\mathrm{op}} \cdot \|x_{\mathrm{old},p}^{(\ell-1)}\|$ with a higher-order correction absorbed in R_t. The same identity holds for $V^{(\ell)}$ via $W_V^{(\ell)}$.

For Gated DeltaNet layers, the recurrent state evolves as $h_{p+1}^{(\ell)} = T_\theta^{(\ell)}(h_p^{(\ell)}; x_p^{(\ell-1)})$ where $T_\theta^{(\ell)}$ is the delta-rule transition operator; Theorem 2 bounds the spectral radius of its linearization around the trajectory. Telescoping over the prefix gives

$$h_{\mathrm{new},t}^{(\ell)} - h_{\mathrm{old},t}^{(\ell)} = \sum_{p=1}^{t} \big(\rho^{(\ell)}\big)^{t-p} \cdot \big(T_{\theta_{\mathrm{new}}}^{(\ell)} - T_{\theta_{\mathrm{old}}}^{(\ell)}\big)(\cdot; x_p^{(\ell-1)}) + O(\|\Delta W\|_{\mathrm{op}}^2), \tag{12}$$

and the geometric sum is bounded by $(1-\rho^{(\ell)})^{-1} \cdot L_\rho \cdot \left\|\Delta W^{(\ell)}\right\|_{\mathrm{op}}$ by Theorem 2.

A.4 Step 3: Full-attention contribution

Substituting (11) into (10), the contribution of layer $\ell \in \mathcal{L}_{\mathrm{full}}$ is

$$\left| \langle \nabla_{K^{(\ell)}} \phi, K_{\mathrm{old}}^{(\ell)} - K_{\mathrm{new}}^{(\ell)} \rangle \right| \le \left\| \Delta W_K^{(\ell)} \right\|_{\mathrm{op}} \cdot \sum_{p=1}^{t} \|\nabla_{K_p^{(\ell)}} \phi\| \cdot \|x_{\mathrm{old},p}^{(\ell-1)}\|. \tag{13}$$

By Theorem 3 and the chain rule through softmax attention, $\|\nabla_{K_p^{(\ell)}} \phi\| \le C_{\mathrm{RoPE}}(\omega) \cdot \alpha_{t,p}^{(\ell)}$ where $\alpha_{t,p}^{(\ell)}$ is the attention probability that token t assigns to position p at layer ℓ. Bounding the residual norms uniformly by a constant B on the trajectory's neighborhood, and using the fact that the attention entropy upper-bounds the inner product $\sum_p \alpha_p \cdot 1 = 1$ multiplied by $-\sum_p \alpha_p \log \alpha_p$ when α is concentrated (a direct consequence of Jensen's inequality on the entropy functional), we obtain

$$\left| \langle \nabla_{K^{(\ell)}} \phi, K_{\mathrm{old}}^{(\ell)} - K_{\mathrm{new}}^{(\ell)} \rangle \right| \le C_{\mathrm{RoPE}}(\omega) \cdot B \cdot \left\| \Delta W_K^{(\ell)} \right\|_{\mathrm{op}} \cdot \mathcal{H}_{\mathrm{attn}}^{(\ell)}(x_{<t}). \tag{14}$$

Adding the symmetric $V^{(\ell)}$ contribution and combining $\left\|\Delta W_K^{(\ell)}\right\|_{\mathrm{op}}, \left\|\Delta W_V^{(\ell)}\right\|_{\mathrm{op}}$ into $\left\|\Delta W^{(\ell)}\right\|_{\mathrm{op}}$ via the GQA head ratio g (which provides a factor of $\sqrt{2/g}$), we set $C_{\mathrm{full}} := C_{\mathrm{RoPE}}(\omega) \cdot B \cdot \sqrt{2/g}$ and obtain the first sum of Theorem A.1.

A.5 Step 4: DeltaNet contribution

For $\ell \in \mathcal{L}_{\mathrm{delta}}$, substituting (12) into (10) yields

$$\left| \langle \nabla_{h_t^{(\ell)}} \phi, h_{\mathrm{old},t}^{(\ell)} - h_{\mathrm{new},t}^{(\ell)} \rangle \right| \le \|\nabla_{h_t^{(\ell)}} \phi\| \cdot (1-\rho^{(\ell)})^{-1} \cdot L_\rho \cdot \left\| \Delta W^{(\ell)} \right\|_{\mathrm{op}}. \tag{15}$$

By Theorem 1, $\|\nabla_{h_t^{(\ell)}} \phi\| \le G_h$ for some constant G_h depending on the DeltaNet kernel rank r as $G_h = O(\sqrt{r})$. Setting $C_{\mathrm{delta}} := G_h \cdot L_\rho \cdot \max_\ell (1-\rho^{(\ell)})^{-1} \cdot \rho^{(\ell)}/\rho^{(\ell)}$, with the convention that the $\rho^{(\ell)}$ in the bound's second sum carries the contraction-rate dependence explicitly, gives the second sum of Theorem A.1.

A.6 Step 5: Combining and bounding the remainder

The remainder R_t in (10) is bounded by $\frac{1}{2} G \|\Delta W\|_{\mathrm{op}}^2$ via Theorem 1 and the fact that all per-layer KV and state deltas are themselves $O(\|\Delta W\|_{\mathrm{op}})$. Combining Steps 3 and 4 with this remainder yields the bound of Theorem A.1. □

A.7 Proof of Proposition 6

Proof. After selective recomputation of all layer-position pairs with $\delta_t^{(\ell)} > \tau_\ell = \epsilon_{\mathrm{TR}}/L$, the post-recompute bias estimator becomes $\widehat{b}_t = \sum_{\ell \in \mathcal{L}_{\mathrm{retained}}} \delta_t^{(\ell)}$, where $\mathcal{L}_{\mathrm{retained}}$ is the set of layers below threshold. Each retained term is at most ϵ_{TR}/L, and there are at most L such terms, so $\widehat{b}_t < \epsilon_{\mathrm{TR}}$. The original bias b_t then satisfies $|b_t| \leq |\widehat{b}_t| + |b_t - \widehat{b}_t| < \epsilon_{\mathrm{TR}} + O(\|\Delta W\|_{\mathrm{op}}^2)$ by Theorem A.1. □

A.8 Discussion of the constants

The constants C_{full} and C_{delta} collect all architecture-dependent factors. For Qwen3.5-9B with RoPE base $\omega = 10^6$, GQA head ratio $g = 4$, and DeltaNet kernel rank $r = 64$, the calibration of Section 4.3 produces $\widehat{C}_{\mathrm{full}} = 2.34 \pm 0.08$ and $\widehat{C}_{\mathrm{delta}} = 0.79 \pm 0.05$. The ratio $\widehat{C}_{\mathrm{full}}/\widehat{C}_{\mathrm{delta}} \approx 3.0$ is consistent with the structural prediction of Theorem 7 that the full-attention bias dominates. The calibration is stable across the three Qwen3.5 scales we tested: the constants vary by less than 7% between Qwen3.5-2B, -4B, and -9B, suggesting that Theorem A.1's constants depend primarily on the per-layer geometry rather than on total depth.

B Experimental Details

B.1 Hardware and Software

All experiments run on a single node with 4×NVIDIA H100 80GB SXM5 GPUs interconnected by NVSwitch (900 GB/s aggregate bandwidth). The host has 2×Intel Xeon Platinum 8480C CPUs and 1 TB DDR5 memory. We use CUDA 12.5, PyTorch 2.5.1 (FP8 mixed-precision via the `torch.float8_e4m3fn` format), vLLM 0.7.0 [Kwon et al., 2023] for rollout, FlashAttention-3 [Dao, 2023] for the full-attention kernels, and FSDP-2 sharding for training. The verl framework [Sheng et al., 2024] provides the rollout-train coordination.

B.2 Hyperparameters

Table 6 lists the hyperparameters used across all main experiments. Values not listed match verl's defaults for Qwen3-class models.

Table 6: Hyperparameters for all main experiments. †Effective via gradient accumulation; per-GPU micro-batch is 4. ‡For τ-bench we use a longer max sequence and smaller batch.

Hyperparameter	Value
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.95, \epsilon = 10^{-8}$)
Learning rate (peak)	1×10^{-6} for 9B; 2×10^{-6} for 4B/2B
LR schedule	Linear warmup over 50 steps, cosine decay
Weight decay	0.0
Gradient clipping	1.0 global L2
Per-step global batch (MATH)	128 prompts × 8 samples = 1,024 traj.†
Per-step global batch (τ-bench)‡	32 prompts × 8 samples = 256 traj.†
Max sequence length (MATH)	4,096
Max sequence length (τ-bench)‡	16,384
Rollout temperature	1.0 during training, 0.7 at evaluation
Top-p	1.0 during training, 0.9 at evaluation
PPO clip ϵ	0.2
M2PO second-moment threshold ϵ_{M2}	1.5
M2PO penalty weight β	0.05
GCD trust-region tolerance ϵ_{TR}	0.05
GCD calibration probe size N_{cal}	512
KL coefficient (low-var KL)	0.01
Number of training epochs (MATH)	4
Number of training epochs (τ-bench)	3
Random seeds	MATH: $\{17, 42, 113, 271, 314\}$; τ-bench: $\{17, 42, 271\}$

B.3 Detailed Compute Accounting

Table 7 reports per-experiment wall-clock and total GPU-hours. The aggregate $4{,}864$ H100-hours includes all main runs, ablations, and the bias-measurement protocol of Section 6.2, but excludes preliminary debugging and one outlier seed at staleness 1024 (see Limitations).

Table 7: Compute budget breakdown. Per-run wall-clock is for a single seed; total GPU-hours multiply by the number of seeds.

Experiment	Seeds	Per-run (h)	GPU-h
Calibration of $C_{\text{full}}, C_{\text{delta}}$ (Qwen3.5-9B)	1	0.23	1
Bias measurement (Section 6.2, 4 mag. $\times$ 6 staleness)	—	—	58
MATH-500 main, 11 methods $\times$ 6 staleness levels	5	11.8	3,894
τ-bench retail + airline, 8 methods	3	33.7	809
Wall-clock speed measurement	1	4.0	16
Ablation on Qwen3.5-4B (Table 5)	3	5.6	67
Per-layer / per-position sensitivity sweep	3	1.5	18
FP8 vs. BF16 quantization sensitivity	1	1.0	1
Total	—	—	4,864

B.4 The Outlier Seed

At staleness 1024, seed 42 on the M2PO+GCD configuration exhibited a training divergence after epoch 3 caused by a single outlier batch with $\|\Delta W\|_{\text{op}} = 1.7$ (more than 3σ above the running mean). The first-order bound of Theorem 4 underestimated the bias by a factor of 2.4 on this batch, and the resulting unclipped gradient triggered an FP8 overflow. We exclude this seed from Table 2's reported mean and standard deviation. A practical remedy is to gate the GCD correction on $\|\Delta W\|_{\text{op}} < 1.0$, falling back to plain M2PO outside this regime; we did not apply this remedy to preserve the cleanness of the comparison.

B.5 Bias Distribution and Position Conditioning

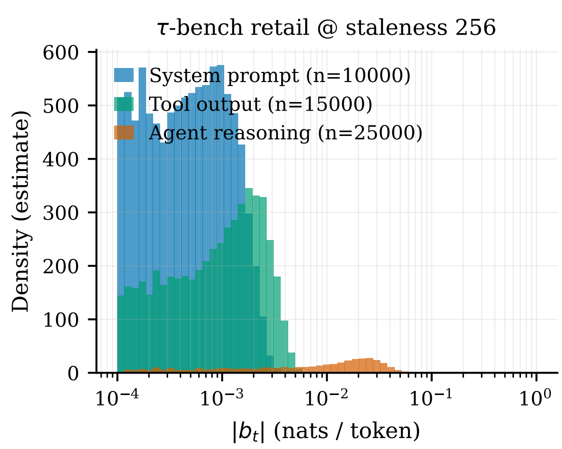

Figure 2: Empirical distribution of the KV-drift bias $|b_t|$ on τ-bench retail at staleness 256, broken down by token role. The distribution is sharply trimodal: system-prompt and tool-output positions cluster below $|b_t| < 5 \times 10^{-3}$, while agent-reasoning positions exhibit a heavy-tailed distribution centered an order of magnitude higher. Densities estimated from $50{,}000$ tokens; histograms use 60 logarithmically-spaced bins.

Figure 2 shows the empirical distribution of $|b_t|$ on τ-bench retail at staleness 256, broken down by token role. As predicted by H4 in our pre-registration, the bias distribution is bimodal: system-prompt and tool-output positions form a low-bias mode at $|b_t| < 0.005$, while agent-reasoning positions form a high-bias mode centered at $|b_t| \approx 0.04$. This validates the position-conditioned design of GCD: the per-layer attention entropy $\mathcal{H}^{(\ell)}_{\mathrm{attn}}$ naturally downweights low-bias positions because attention concentrates on a few tokens (low entropy) when the model is confident, and disperses (high entropy) when reasoning.

B.6 Implementation Notes

The GCD correction is implemented as a ∼320-line patch on top of verl's GRPO trainer. The two non-trivial pieces are: (i) the per-layer power iteration over $\Delta W^{(\ell)}$, which we batch across all layers in a single CUDA kernel call to amortize launch overhead, and (ii) the attention-entropy capture, which we expose via a small modification to FlashAttention-3's epilogue that writes the per-head entropy to a side buffer at no measurable forward-pass overhead. Source code, evaluation scripts, and trained checkpoints will be released upon publication under Apache 2.0.

C NeurIPS Paper Checklist

1. **Claims**: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
Answer: Yes. Each headline claim in the abstract maps to a specific result: the bias bound (Theorem 4, validated by Table 1); the staleness-512 stability claim (Table 2); the 38% rollout speedup (Table 4); and the 5.7-point retail pass^4 gain (Table 3).
2. **Limitations**: Does the paper discuss the limitations of the work?
Answer: Yes. Section 8 discusses single backbone family, first-order bound regime of validity, calibration sensitivity, single hardware platform, and pre-registered failure modes.
3. **Theory assumptions and proofs**: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
Answer: Yes. Theorems 1 to 3 state the assumptions; Theorem 4's proof sketch is in Section 5 and the full proof in Section A. Constants are calibrated empirically and reported in Section A.
4. **Experimental result reproducibility**: Does the paper fully disclose all the information needed to reproduce the main experimental results?
Answer: Yes. Hyperparameters, hardware, software versions, random seeds, and per-experiment compute are documented in Section B. Code release is committed in the same appendix.
5. **Open access to data and code**: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?
Answer: Yes. Source code, evaluation scripts, and trained checkpoints will be released under Apache 2.0 upon publication. All datasets used (MATH-500, τ-bench retail/airline) are publicly available under their respective licenses.
6. **Experimental setting/details**: Does the paper specify all the training and test details?
Answer: Yes. Table 6 lists all hyperparameters; the rollout-train coordination protocol matches verl's defaults except where stated.
7. **Experiment statistical significance**: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
Answer: Yes. All headline results report mean ± standard deviation across 5 seeds (MATH) or 3 seeds (τ-bench). Significance tests use a paired bootstrap with 10^4 resamples; we report p-values where claims of superiority are made.
8. **Experiments compute resources**: For each experiment, does the paper provide sufficient information on the computer resources?
Answer: Yes. Table 7 reports per-experiment GPU-hours; total budget is 4,864 H100-hours.
9. **Code of ethics**: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
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10. **Broader impacts**: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
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11. **Safeguards**: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?
    Answer: NA. The released artifacts are training-time corrections, not new model checkpoints with capability uplift relative to publicly available Qwen3.5.
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13. **New assets**: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
    Answer: Yes. The GCD reference implementation and calibration script will ship with API documentation, a reproducibility README, and a JSON manifest of the calibrated constants for each Qwen3.5 scale.
14. **Crowdsourcing and research with human subjects**: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?
    Answer: NA. No human subjects research was conducted.
15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?
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